

Week 1: Introduction to Full Stack & MERN Stack

1. Overview of Full Stack Development & Role of a Developer

Full Stack Development refers to building both the frontend and backend of a web application. Frontend is the part users interact with, while backend manages server operations, business logic, and databases. A full stack developer is responsible for handling the complete workflow of an application, from designing interfaces to managing data storage and deployment. This role requires knowledge of multiple technologies and problem-solving skills to build efficient and scalable applications.

2. Introduction to MERN Components (MongoDB, Express, React, Node.js)

The MERN Stack is a popular technology stack used for full stack web development because all its components use JavaScript. MongoDB is a NoSQL database used to store flexible and structured data. Express.js is a lightweight backend framework that simplifies API creation and routing.

React.js is a frontend library used to build dynamic and reusable user interfaces. Node.js is a runtime environment that allows JavaScript to run on the server side. Together, these technologies make web development faster, organized, and scalable.

3. Installing Node.js, npm, VS Code

Node.js is required to run JavaScript applications outside the browser. Installing Node.js also installs npm, which is used to manage libraries, dependencies, and packages for projects. npm helps developers save time by easily adding tools and frameworks. VS Code (Visual Studio Code) is a modern code editor with features like syntax highlighting, debugging, extensions, and terminal support. Setting up these tools creates a strong development environment for coding and testing applications.

4. Setting up MongoDB Locally & MongoDB Compass

MongoDB is a NoSQL database that stores data in a document-based format instead of tables. Setting up MongoDB locally helps developers test and manage databases on their own system before deployment. It is useful for learning database creation, collections, and CRUD operations. MongoDB Compass is a graphical user interface that allows developers to visually interact with databases, create collections, run queries, and analyze stored data. This makes database handling simpler and more user-friendly.

5. npm init & Basic Terminal Commands

The npm init command is used to create a package.json file, which contains important project information such as dependencies, scripts, and version details. It is the first step in

setting up a Node.js project. Basic terminal commands such as `cd`, `mkdir`, `ls`, and `pwd` are essential for navigating and organizing files efficiently. Learning terminal commands improves speed, productivity, and confidence in development.

6. Git/GitHub Setup

Git is a distributed version control system that helps developers track changes in code and manage different project versions. It allows safe experimentation without losing previous work. GitHub is a cloud-based platform used to store repositories and collaborate with other developers. Setting up Git and GitHub allows version tracking, project backup, teamwork, and easy deployment. Common commands like `git init`, `git add`, `git commit`, and `git push` are important for maintaining project history and collaboration.

Conclusion

This week provided a strong introduction to Full Stack Development and the MERN Stack. By learning core technologies, setting up development tools, understanding databases, and using version control systems, a solid foundation was built for future MERN stack projects. These concepts are essential for developing modern, scalable, and interactive web applications.